PABLO **YESTE BLESA**

Junior engineer & Mathematics Student

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PROFILE

MSc student in Mathematics & Engineering at TUM with a background in **optimization, control, and machine learning.** Experienced in **trajectory optimization, motion planning, and surrogate modeling** within industrial R&D. Co-developer of NLOTrajectories, a Siemens project integrating neural network-based signed distance fields into nonlinear optimization pipelines. Motivated to apply advanced mathematics to real-world challenges in robotics and energy systems.

EXPERIENCE

2025 (Present)
Research Project - Siemens AG & TUM

Trajectory Optimization in Learned Environments

public repo: github.com/pyMSE/NLOTrajectories

- Developed a **nonlinear trajectory optimization pipeline** (CasADi, IPOPT) **integrating NN-based** signed distance fields.
- Designed **benchmarks for multiple dynamics** (point-mass, unicycle, Ackermann) in convex and non-convex environments.
- Evaluated solver performance using IoU, Chamfer, and Hausdorff metrics, **comparing learned vs. classical SDFs**.
- Demonstrated **scalability** and potential for **integration in industrial environments**.

2023 - 2025 (present)
Universidad Politécnica de Madrid (UPM)

Junior Engineer Researcher

- Co-authored Towards certification... (Expert Systems with Applications, 2025) through the implementation of an **industrial validation pipeline for surrogate models** with Airbus.
- Developed **aerodynamic surrogate models** (PODI, MP-GNN) for surface pressure prediction, achieving results consistent with DLR benchmarks and outperforming baseline methods (CETACEO, NextGenEU).
- Built a **modular, scalable Python framework** with advanced UI; released in **pyLowOrder** (https://github.com/ArnauMiro/pyLowOrder)
- Engineered **geometry preprocessing routines** for RANS meshes (~1.7M nodes), reducing computation from hours to seconds.
- Currently validating surrogate models on multi-GPU (MareNostrum, BSC) and co-authoring a comparative study of **MP-GNN, KAN, and MLP models**, with MP-GNN achieving superior performance (TIFON, NextGenEU).

AWARDS & PROJECTS

- Biologically Inspired Learning for Humanoid Robots (6 ECTS, grade: 1.3). Lab course at TUM on **ROS2-based humanoid robotics**, applying **reinforcement learning** and biologically inspired ML algorithms.
- Excellence Scholarship, Comunidad de Madrid (2019). Awarded for outstanding academic performance in Aerospace Engineering.

EDUCATION

2024 - present (4rd semester)
Technische Universität München
MS Mathematics in Sci. and Eng.
Selected coursework: Optimization, Robotics, Machine Learning

2019 - 2024
Universidad Politécnica de Madrid
BS Aerospace Engineering

TECHNICAL SNAPSHOT

Python · PyTorch · CasADi · IPOPT · Graph Neural Networks · ROS2 · Reinforcement Learning

LANGUAGES

- English (full professional)
- German (conversational)
- Spanish (native)
- French (conversational)
- Italian (conversational)